Balaban through Newton's polygons: a 0+1 lesson

Lesson and written construction, 2026-10-27. A useful way to approach Balaban's method is to start with a familiar existence problem: construct motion from finitely many force impulses, prove estimates that survive arbitrary insertion of intermediate times, and take a limit. Then ask what additional description must survive when an intermediate position is integrated rather than solved for. The answer includes its fluctuation law, normalization and generated interactions.

We work with one mechanical coordinate and time, keeping $\hbar$ explicit. The finite construction is a modern use of Newton's polygon idea; it is not a proof about every historical edition or force polygon. The Galileo inertial-line / falling-parabola comparison supplies our action observable. No relativistic assumption or c is needed.

The ODE proof below is standard in method and claims no novelty. Its purpose is to make the comparison concrete, including arbitrary unequal time steps and either force-kick convention. The two retained-weight calculations are in the time-blocking note and the nonlinear cell. They specify their proof status and source reading. Balaban's ICM overview, printed pages 1260–1262 (text read), and Dimock I, sections 2.1, 2.4, 4.5 and Theorem 14 (selected passages read), motivate the comparison.

Targeted check (GPT-6.1 Terra, 2026-10-27). Terra read the lesson and checked the causal stability, residual, Cauchy and limit arguments in sections 1–3 without finding a mathematical error under their global Lipschitz hypothesis. The lesson's distinction between a trajectory and a weighted law was also checked. This was a targeted written review; the physical origin of the weighting is still open.

1. Start with a finite Newton polygon

Let $m > 0$ be the mass and $f(q) = F(q)/m$ the acceleration. The desired motion would satisfy

$$\dot{q} = v, \qquad \dot{v} = f(q), \qquad (q(0), v(0)) = (q_0, v_0). \tag{1}$$

For now we do not assume that a solution exists. Given a positive time step τ, define a drift and a force kick by

$$D_\tau(q, v) = (q + \tau v, v), \qquad K_\tau(q, v) = (q, v + \tau f(q)). \tag{2}$$

A step can kick first, $D_\tau K_\tau$, or drift first, $K_\tau D_\tau$. The first choice uses the force at the old position; the second uses the force at the new position. Both give a finite piecewise straight trajectory with impulses. Neither is assumed to agree exactly with the unknown continuous motion.

Fix $T > 0$, take any finite positive partition of $[0, T]$, choose either ordering in each cell, and compose the maps chronologically. There is no equal-step or nesting requirement. When a time is inserted, recompute the later states from the fixed initial state and the prescribed force rule.

Existence of the limit requires two kinds of estimates. Stability bounds how differences grow. Consistency bounds the error in the finite force balance. The two together will construct the ODE solution.

2. Lipschitz control supplies stability

Assume for this construction that

$$|f(q) - f(Q)| \le \Lambda |q - Q| \quad (q, Q \in \mathbb{R}), \qquad \Lambda \ge 0. \tag{3}$$

The quantity Λ has units of inverse time squared. Choose a reference frequency $\rho > 0$ and use the length-valued phase-space norm

$$\|(q,v)\|_\rho = |q| + \frac{|v|}{\rho}, \qquad \alpha = \max(\rho, \Lambda/\rho), \quad \beta = \rho + \Lambda/\rho, \quad b = |f(0)|/\rho. \tag{4}$$

There is no action constant in these definitions. Directly from (2),

$$\mathrm{Lip}(D_\tau) \le 1 + \rho\tau, \qquad \mathrm{Lip}(K_\tau) \le 1 + \Lambda\tau/\rho. \tag{5}$$

Thus either cell map has Lipschitz constant at most $e^{\beta\tau}$. The constants multiply under composition, and their exponents add. Every finite polygon therefore has the same upper amplification $e^{\beta T}$, independently of the number and placement of vertices.

We also need a bound on the states themselves, because consistency involves the force along the polygon. For either cell map M_τ,

$$\|M_\tau z\|_\rho \le e^{\beta\tau}\|z\|_\rho + \tau e^{\beta\tau} b. \tag{6}$$

For a kick-first cell, the inhomogeneous term is actually at most $\tau(1+\rho\tau)b$; for a drift-first cell it is τb. Iterating (6) gives the common vertex bound

$$R = e^{\beta T}(\|z_0\|_\rho + Tb). \tag{7}$$

In particular $|q_j| \le R$, $|v_j| \le \rho R$, and $|f(q_j)| \le A := |f(0)| + \Lambda R$. A very irregular partition has not created a new force or stability constant.

3. The polygons construct the ODE solution

Linearly interpolate both vertex coordinates to obtain $z_\pi(t) = (q_\pi(t), v_\pi(t))$. The displayed velocity is continuous; it differs from the physical constant velocity along each straight polygon segment by at most that cell's kick. This distinction disappears with the mesh.

Let $\delta = |\pi|$ be the maximum cell duration. In a kick-first cell, with elapsed time $s \in [0, \tau]$,

$$\dot q_\pi - v_\pi = (\tau - s) f(q_j), \qquad \dot v_\pi - f(q_\pi) = f(q_j) - f(q_\pi). \tag{8}$$

In a drift-first cell the analogous expressions are $-sf(q_{j+1})$ and $f(q_{j+1}) - f(q_\pi)$. The bounds in (7) give, for almost every time,

$$\|\dot z_\pi - \mathcal{F}(z_\pi)\|_\rho \le C\delta, \quad \mathcal{F}(q,v) = (v, f(q)), \quad C = |f(0)| + 2\Lambda R. \tag{9}$$

The vector field $\mathcal{F}$ is Lipschitz with constant α in the norm (4). Two polygons with the same initial state consequently satisfy

$$\sup_{t \le T} \|z_\pi(t) - z_\sigma(t)\|_\rho \le C \frac{e^{\alpha T} - 1}{\alpha} (|\pi| + |\sigma|). \tag{10}$$

To verify this estimate, integrate the two residual equations (9), subtract, and apply the elementary integral Gronwall inequality. That inequality follows by integrating the differential inequality for an upper bound after multiplying by $e^{-\alpha t}$.

Completeness of continuous path space now constructs a uniform limit $z(t)$. The residual integral tends to zero; the Lipschitz bound passes $\mathcal{F}(z_\pi)$ to $\mathcal{F}(z)$ under the time integral. Hence

$$z(t) = z_0 + \int_0^t \mathcal{F}(z(s))\, ds. \tag{11}$$

The limit is differentiable and solves (1). Applying the same difference estimate with zero residual proves uniqueness. Letting one mesh tend to zero in (10) gives a quantitative error for any other partition. Both kick orders, mixed arbitrarily, lead to the same solution. Since T was arbitrary, the global Lipschitz hypothesis constructs forward motion for every finite time.

This is the first analogy with Balaban: one proves bounds for each finite description before appealing to a limiting object. The limit follows from estimates uniform in the refinement. Nothing in this argument requires choosing $\hbar > 0$.

4. Local existence, continuation and endpoint caustics are different

For a locally Lipschitz force one first works inside a bounded region. A Lipschitz extension permits the construction above; a short-time bound ensuring that the trajectory remains in the region makes the extension irrelevant. Continuation requires additional control against escape from all bounded phase-space regions.

For example, $V(q) = \lambda q^4/4$, $\lambda > 0$, has unbounded force derivative globally. A fixed energy bounds both q and $p = mv$, so its initial-value solution continues. The inverted quartic has a smooth, locally Lipschitz force too, yet admits finite-time escape. The explicit bounds and escape solution appear in the time-blocking note, section 6. Continuity alone can give local existence; the Lipschitz assumption above also supplies the uniqueness and quantitative stability used in (10).

An endpoint problem fixes $q(0)$ and $q(T)$ rather than initial velocity. It has a separate possible singularity. For the harmonic oscillator the full phase-space derivative is

$$\begin{pmatrix} \cos \omega T & \sin(\omega T)/\omega \\ -\omega \sin(\omega T) & \cos \omega T \end{pmatrix}, \qquad \det = 1. \tag{12}$$

The position response to initial velocity is $\sin(\omega T)/\omega$. It vanishes at a conjugate time while the full phase-space map stays invertible. The corresponding endpoint action Hessian can fail to invert even though the initial-value motion remains unique. The time-blocking note gives a finite-polygon determinant that approaches this singularity under refinement.

Therefore a failure of one position-only generating function is not, by itself, a failure of Newton's dynamics. A physical completion argument has to establish which extra information the complete theory needs and why it cannot be retained classically.

5. Retain an intermediate time as a fluctuating variable

The ODE polygon solves for its intermediate states. A different construction assigns weights and integrates an unobserved intermediate position. For the positive free kernel,

$$k_{T,\hbar}(y, x) = \sqrt{\frac{m}{2\pi\hbar T}} \exp\left[-\frac{m(y-x)^2}{2\hbar T}\right], \qquad \hbar > 0, \tag{13}$$

splitting $T = u + v$ gives a conditional inserted position with

$$\mu = \frac{vx + uy}{T}, \qquad \mathrm{Var}(Z \mid x, y) = \frac{\hbar uv}{mT}. \tag{14}$$

A constant-force Newton parabola shifts this vertex from its chord by $-Fuv/(2m)$. The associated hat deformation has action $F^2uvT/(8m)$. For equal-noise Gaussian bridge laws the relative entropy between the shifted and unshifted vertex laws is that action divided by $\hbar$. The precise normalization and physical premises are recorded in the time-blocking note, equations (15a)–(15b).

The positive Gaussian kernel is a Euclidean construction. The real-time weight uses $e^{iS^R/\hbar}$ and Fresnel integration; its quadratic second moment is not a probability covariance. Changing between them requires more than replacing a symbol in a fluctuation estimate.

Here the role of the action constant is visible: it sets the fluctuation width and the strength of action differences in the weight. Ordinary ODE convergence controls the mean path construction; it does not fix that extra law.

6. What a nonlinear time block actually produces

For a symmetric two-cell Euclidean action, put $a = mT/(uv)$ and $w = T/2$. Retain

$$A_\hbar(\mu) = -\hbar \log \left[\sqrt{\frac{a}{2\pi\hbar}} \int e^{-[a(z-\mu)^2/2 + wV(z)]/\hbar} dz \right]. \tag{15}$$

This whole function accompanies the coarse kinetic term and the endpoint potential terms. Its derivatives have a useful interpretation:

$$A'_\hbar = a(\mu - \mathbb{E}Z), \qquad A''_\hbar = a - \frac{a^2}{\hbar} \operatorname{Var}(Z). \tag{16}$$

Thus an eliminated fluctuation changes the retained response. For convex V, the nonlinear-cell note proves $0 \le A''_\hbar \le a$ and an explicit normalized tail bound about the conditional minimizing position. It also proves a simultaneous bound for large fluctuations at nonadjacent sites in one halving. No bad region is dropped from the integral.

Classical minimization gives the envelope $A_0(\mu) = \min_z [a(z-\mu)^2/2 + wV(z)]$. It too has curvature between zero and a. Coarse regularity alone therefore does not distinguish the two operations.

The quartic centre does distinguish them quantitatively. For $V(z) = \lambda z^4/4$, $\lambda > 0$, $A''_0(0) = 0$ while

$$A''_\hbar(0) > 0 \quad (\hbar > 0), \qquad A''_\hbar(0) \sim \frac{3w\lambda\hbar}{a} \quad \left(\frac{w\lambda\hbar}{a^2} \downarrow 0 \right). \tag{17}$$

The nonlinear-cell note gives finite inequalities, not just the asymptotic formula, and inverts them to a model-conditional action interval. This is an example of a generated interaction: a purely quartic bare potential acquires quadratic curvature after fluctuation integration. The action multiplier remains supplied in (15).

7. The part of Balaban's method visible here

The correspondence can now be stated concretely.

Mechanical construction	Controlled quantity	What is retained
Force polygon on a time partition	Lipschitz amplification and force-balance residual	Initial state, force rule, and a uniform error estimate
Endpoint stationary elimination	Inverse action Hessian on a specified branch	Effective action and its source response
Gaussian integration	Covariance and determinant	Schur-complement action together with normalization and explicit $\hbar$
Nonlinear integration	Full conditional law and large-field probabilities	Generated interactions, normalization and the regions outside the local approximation

Balaban organizes successive effective densities so that such information can be estimated at the next scale. The analogous task for ODE existence is to preserve enough stability and consistency information to pass from all finite polygons to one motion. The objects of the two existence problems differ: an ODE solution is a trajectory; a weighted continuum construction includes a law or propagator over trajectories.

The one-dimensional chain makes an elementary first step possible: conditioned on retained times, distinct missing intervals are independent. Their interactions can nevertheless change

functional form at each elimination. Closure of a finite list of polynomial coefficients is an additional assertion that must be checked.

8. Consequence for STATE

The requested $0 + 1$/ODE lesson is developed together with a causal construction of nonlinear initial-value motion on arbitrary unequal partitions, allowing mixed kick orders. It complements the endpoint Green-function calculation and explains the normalized nonlinear cell. The research question is now sharper: which independent physical law selects the weight and its action multiplier, beyond the regularity already available from classical motion? The next candidate test is branch competition in a nonlinear retained interaction, keeping the full initial-value description as its classical control. These are written derivations and a pedagogical comparison, not an independent proof of action necessity.